

JU SeqWorkbench Alpha - Quick Start

Language: 한국어 | English

JU SeqWorkbench Alpha is a local desktop application for reviewing and editing small-to-medium Sanger, FASTA, and MSA sequence datasets, grouping sequences, and examining site- or region-centered patterns. It is an alpha testing build, not a clinical diagnostic system or an NGS analysis pipeline.

1. Start the application

1. Open the distribution folder.
2. Run `JU SeqWorkbench Alpha/JU SeqWorkbench Alpha.exe`.
3. Wait for the startup splash to close and the launcher to appear.

Keep backups of important original files and recheck important results against the source data.

2. Open or import sequence data

- Use **File > Open** to open FASTA/TXT, AB1/ABI, or a saved JU SeqWorkbench project in a new viewer.
- Use **File > Import** in an existing viewer to append supported sequence data to that workspace.
- The included `Example_Data/JU_SeqWorkbench_Example_35seq.fasta` can be used for a first test. It is demonstration data and must not be treated as a biological reference.

After loading data, review the Sequence IDs and sequence rows. In Edit mode, the viewer supports focused sequence and ID edits, copy/paste, and the existing undo/redo commands. ID-panel and sequence-body selections are separate interaction contexts, so confirm the active panel before editing.

3. Try Point Visualization

1. Open a sequence dataset in the viewer.
2. Choose **Analysis > Visualization > Point Visualization**.
3. Select the applicable NT, AA, or codon mode and enter one or more valid positions, such as `10,25,50` when they exist in the loaded alignment.
4. Review the result table, warnings, and plot before exporting.

4. Try Region Visualization

1. Choose **Analysis > Visualization > Region Visualization**.
2. Enter a valid continuous region or the ranges required by the selected view.
3. Select the metric and display options appropriate to the loaded data.
4. Run the analysis and review the result table or figure.

Large alignments or color-heavy views can take longer to render. A requested position or range must exist in the available sequence data.

5. Project Save and Export are different

- **Save Project / Save Project As** stores the current JU SeqWorkbench working session as a `.dvproj` project so it can be reopened for continued work.
- **Export** writes a result table, figure, FASTA, or other analysis output for use outside the project.

Saving a project does not replace an explicit data or result export. Undo history from the previous session is not restored when a project is reopened.

6. External alignment tools

MAFFT and Clustal Omega executables are not bundled or downloaded. To run an external MSA, install the tool separately and select its local executable in the application settings. Temporary FASTA files are passed to the selected local program.

7. Feedback

- Bug reports and feature requests: GitHub Issues
- Questions, ideas, and general feedback: GitHub Discussions

For full feature documentation, see `JU_SeqWorkbench_User_Guide_EN.pdf` in the distribution documentation folder.